DATASET DESCRIPTION

The dataset used in this study consists of daily closing prices of multiple US equities over a specified time period. Each row represents a trading day, while each column corresponds to a different stock. The "Date" column records the observation date, and the remaining columns contain the respective stock prices.

This structure provides a time series dataset that enables the analysis of stock performance over time. However, raw price data alone is not directly suitable for risk–return analysis, as stock prices vary significantly in scale across different equities.

To ensure meaningful comparison across stocks, the price data was transformed into **returns**. Returns standardize the data and reflect the relative change in price rather than absolute price levels.

DATA PREPROCESSING

The daily return for each stock was computed using the formula:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where P_t represents the price at time t, and P_{t-1} represents the price at the previous time period.

This transformation is essential because:

- It allows comparison between stocks with different price ranges
- It reflects actual investment performance
- It forms the basis for risk and volatility analysis

In addition, return-based data is widely used in financial analysis as it provides a more accurate representation of market behavior.

CORRELATION ANALYSIS

A correlation matrix was constructed to examine the relationships between the returns of different stocks. Correlation measures the degree to which two stocks move in relation to each other.

- A correlation close to **+1** indicates that stocks move together
- A correlation close to **0** indicates little to no relationship
- A correlation close to **–1** indicates that stocks move in opposite directions

This analysis is important for identifying diversification opportunities. Stocks that are weakly correlated or negatively correlated can be combined to reduce overall portfolio risk.

DESCRIPTIVE STATISTICS

Descriptive statistics were calculated for the return series of each stock to summarize their performance characteristics. The key measures include:

- **Mean return**: Represents the average return of a stock over the observed period and serves as an estimate of expected return.
- **Standard deviation**: Measures the variability of returns and is used as an indicator of risk or volatility.

Stocks with higher mean returns are generally more profitable, while those with higher standard deviations exhibit greater price fluctuations and are considered riskier.

These statistics provide a quantitative foundation for comparing the performance and risk profiles of different stocks.

Risk–Return Scatter Plot

A scatter plot of mean return against standard deviation was constructed for all stocks. This visualization illustrates the trade-off between risk and return.

Stocks located in the upper-right region of the plot represent **high-risk, high-return** investments, while those in the lower-left region represent **low-risk, low-return** investments. This plot serves as a key tool for comparing stock performance.

Contribution to Research Question

This data exploration stage establishes a fundamental understanding of the **risk–return characteristics of individual stocks**. By transforming price data into returns, computing statistical measures, and analyzing relationships between stocks, this section provides the necessary groundwork for subsequent analysis.

The insights gained here are essential for evaluating investment decisions and support further investigation into portfolio construction and optimization.